Subject Code:MATH1065	ENGINEERING MATHEMATICS-II	L	T	P	C
Version 1.0		3	1	0	4
Pre-requisites/Exposure	Basic Mathematics (10+2 level), Advanced Engineering Mathematics-I				
Co-requisites/Exposure	–				

Course Objectives

The aim of the course is to prepare the students to understand and appreciate the power of mathematics as a unifying language transcending a variety of engineering and science disciplines. The focus for the designing of the syllabus has been to provide the students with insights into the mathematical concepts and their applications without much compromising mathematical rigor.

Course Outcomes

On completion of this course, the students will be able to

CO1. Formulate appropriate numerical and optimization schemes for the development of computational algorithms in science and engineering.
CO2. Understand the fundamental tools of complex analysis for addressing and solving a variety of problems viz. special functions, integral transforms, and PDEs.
CO3. Identify and illustrate the use of Bessel functions and Legendre polynomials in real world application.
CO4. Demonstrate computational implementation of integral transforms and their applications.
CO5. Model real-world phenomena evolving in space and time governed by linear PDEs.

Course Description

A good grounding in mathematical sciences has become necessary for the emerging areas like AI/ML, Big data analytics, Underwater image processing, Quantum computing, Cybersecurity, Computational biology, etc. Engineering Mathematics II is intended to improve students' computational and analytical abilities, enabling them to model and create algorithms for a range of real-world problems. To this end, a number of topics are covered, including numerical methods, optimization, complex analysis, integral transforms, special functions, and linear PDEs. It will be beneficial for students to have knowledge of gradient search algorithms as machine learning and many other areas involve some kind of gradient search in optimization. Further, it may be emphasized that complex analysis is useful in several branches of science and engineering such as probability theory, number theory, thermodynamics, signal processing, and image processing. Realizing the importance of partial differential equations and their visualization across many disciplines, a brief introduction to linear PDEs have become prerequisites to gaining insight into the evolutionary dynamics of realistic problems.

Course Content

Unit 0: Motivation 01 Lecture Hours

Why study this course- Relevance and Significance?

Unit I: Numerical Methods and Optimization **12 Lecture Hours**

Bisection and Newton-Raphson methods, Gauss Elimination and Gauss-Seidel methods, Finite difference operators, Interpolation with equal and unequal intervals, Numerical differentiation, and integration, Numerical solution of ODEs: Picard's method, Euler's method, Runge-Kutta fourth order method.

Introduction to optimization, The Simplex method, Duality, Lagrange multipliers, Convex sets and functions, Elements of Gradient search algorithms: Steepest descent, Newton and Jacobi algorithms, Least Squares method, Application: The Markowitz Model, and Overview of constrained optimization, Hill climbing, Single variable search. Recap of Unit I.

Unit II: Infinite Series and Introduction to Complex Analysis **12 Lecture Hours**

Sequence and series, Convergence tests: p-series, Comparison, Ratio and root test, Alternating series.

Complex number system, Euler's formula, Functions of a complex variable, Hyperbolic functions, Limit and Continuity, Derivative and Analytic functions, Holomorphic functions, Cauchy-Riemann equations, Harmonic functions, Line integral and independence of path, Cauchy's theorem, Cauchy's integral formula, Zeros and singularities of a function, Power series: Taylor's and Laurent's series. Some applications. Recap of Unit II.

Unit III: Introduction to Special Functions **04 Lecture Hours**

Introduction to Power series method, Legendre's equation and Legendre polynomials, Bessel's equation and Bessel functions. Application of Bessel functions: CV Raman's model of Indian drums. Recap of Unit III.

Unit IV: Integral Transforms **08 Lecture Hours**

Laplace Transform and its properties, Shifting Theorems, Laplace Transform of derivatives, integrals, and periodic functions, Heaviside and Dirac Delta Functions. Inverse Laplace transforms, Convolution, Solutions of differential equations using Laplace transforms. Fourier series and applications, Dirichlet's condition, Fourier Transforms, Fourier sine and cosine transforms, Properties of Fourier Transforms, Fast Fourier Transform, Inverse Fourier transforms. Recap of Unit IV.

Unit V: Introduction to PDEs and Applications **08 Lecture Hours**

Introduction to Partial differential equations (PDE) and real-world applications, Classification of PDEs: Elliptic, Hyperbolic, Parabolic, Solution of homogeneous and non-homogeneous linear PDEs, Method of separation of variables using Fourier series, Solution of Heat conduction or Diffusion equation, Connection between diffusion and randomness, Wave Equation, Laplace Equation, and Poisson Equation. Some applications: Air pollution, Traffic model. Recap of Unit V.

Text Books:

1. Kreyszig E., Advanced Engineering Mathematics, Wiley Publications. ISBN: 9780471488859, 2005.
2. James, G. and Dyke, P. P., Advanced Modern Engineering Mathematics. Pearson Education. ISBN: 9781292174341, 2018.
3. Corriou, J. P., Numerical Methods and Optimization: Theory and Practice for Engineers. Springer. ISBN: 9783030893651, 2021.

Reference Books:

1. Zill, D.G. and Shanahan, P.D., Complex analysis. Jones & Bartlett Learning. ISBN: 9789384323127, 2015.
2. Strauss, W.A., Partial Differential Equations - An Introduction. John Wiley & Sons Inc. ISBN: 9780470054567, 2008.

3. Burstein, L. PDE Toolbox Primer for Engineering Applications with MATLAB® Basics. CRC Press, 2022.
4. Goodfellow, I., Bengio, Y., and Courville, A. Deep learning. MIT Press. ISBN: 9780262035613, 2016. (Sections 4.3, 4.4, and 4.5)
5. Chandra, S., Jayadeva, and Mehra, A., Numerical optimization with applications. Narosa. ISBN: 9788173198540, 2009.

Web Resources and Supplementary Materials:

1. Gaudet, S., Gauthier, C., & Léger, S. (2006). The evolution of harmonic Indian musical drums: A mathematical perspective. Journal of sound and vibration, 291(1-2), 388-394.
2. MIT OCW on Complex Variables with Applications. https://ocw.mit.edu/courses/18-04-complex-variables-with-applications-spring-2018/
3. Balakrishnan, V., Selected Topics in Mathematical Physics https://nptel.ac.in/courses/115106086.
4. Hsu T.R., Applied Engineering Analysis, John Wiley & Sons. ISBN: 9781119071181, 2017.
5. Neunzert H., Industrial Mathematics and India: A View from the Outside, TMCB, https://www.themathconsortium.in/publication/timcbulletin/vol2/issue1

Future Scope of Enquiry:

Students may like to explore nonlinear PDE:

i) Reaction-Diffusion equations see Murray J.D., Mathematical Biology: I & II, Springer.
ii) Dissipation and displacement of hotspots in reaction-diffusion models of crime, PNAS, 2010. https://www.pnas.org/doi/epdf/10.1073/pnas.0910921107